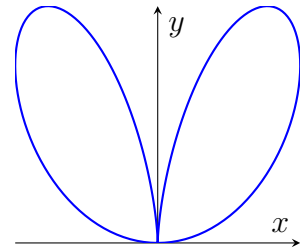


**2025-2026 Spring**  
**MAT124 Midterm**  
**17/04/2026**

1. A point is given in cylindrical coordinates as  $(r, \theta, z) = (\sqrt{3}, -\pi/4, -1)$ . What are its spherical coordinates  $(\rho, \theta, \phi)$ ?

(A)  $(2, -\pi/4, 2\pi/3)$  (B)  $(2, \pi/4, \pi/3)$  (C)  $(4, -\pi/4, 5\pi/6)$   
(D)  $(2, -\pi/4, \pi/6)$  (E)  $(\sqrt{2}, -\pi/4, 3\pi/4)$

2. Which of the following polar equations represents a curve that has loops oriented primarily along the y-axis, as shown in the figure?



(A)  $r = \cos^2 \theta \sin \theta$  (B)  $r = \sin^2 \theta \cos \theta$  (C)  $r = 1 + \sin \theta$   
(D)  $r = \sin(2\theta)$  (E)  $r = 3 \cos \theta$

3. Find the projection of the point  $P(1, 2, 3)$  onto the plane  $x + y + z = 0$ .

(A)  $(-1, 0, 1)$  (B)  $(1, 1, -2)$  (C)  $(0, 0, 0)$  (D)  $(-1, -1, 2)$  (E)  $(1, 2, -3)$

4. Find the area of the region inside the circle  $r = 3 \cos \theta$  and outside the cardioid  $r = 1 + \cos \theta$ .

(A)  $\pi$  (B)  $\pi/2$  (C)  $2\pi$  (D)  $3\pi/4$  (E) 1

5. Identify the quadric surface defined by the equation  $2z^2 - x^2/2 - y^2 - 5 = 0$ .

(A) Hyperboloid of two sheets (B) Hyperboloid of one sheet  
(C) Elliptic paraboloid (D) Cone (E) Cylinder

6. In which direction does  $f(x, y, z) = x^2 + 2y^2 - 3z$  increase most rapidly at  $P(1, 1, 2)$ ?

(A)  $\langle 2, 4, -3 \rangle$  (B)  $\langle 1, 1, -3 \rangle$  (C)  $\langle -2, -4, 3 \rangle$  (D)  $\langle 2, 4, 0 \rangle$  (E)  $\langle 1, 2, -3 \rangle$

7. Let  $w = \ln(x^2 + y^2)$  where  $x = s \cos t$  and  $y = s + \sin t$ . Find  $\frac{\partial w}{\partial t}$  at  $(s, t) = (1, \pi/6)$ .

(A) 0 (B) 1 (C)  $3/2$  (D)  $\sqrt{3}/2$  (E)  $\sqrt{3}/3$

8. Find the arc length of the curve  $y = x^2/2, z = x^3/6$  from  $x = 0$  to  $x = 2$ .

(A)  $10/3$  (B)  $8/3$  (C) 4 (D)  $7/3$  (E) 3

9. A vector parallel to the tangent line of the intersection of  $z = x^2 + 2y^2$  and  $x^2 + (y-1)^2 = 1$  at  $P(1, 1, 3)$  is:

(A)  $\langle 0, 2, 8 \rangle$  (B)  $\langle 2, 4, -1 \rangle$  (C)  $\langle 2, 0, 0 \rangle$  (D)  $\langle 1, 1, 3 \rangle$  (E)  $\langle 4, -2, -8 \rangle$

10. Which is a valid parametrization for the intersection of  $z = x^2 + y^2$  and  $z = 4$ ?

(A)  $\vec{r}(t) = \langle 2 \cos t, 2 \sin t, 4 \rangle$  (B)  $\vec{r}(t) = \langle \cos t, \sin t, 4 \rangle$   
(C)  $\vec{r}(t) = \langle 2 \cos t, 2 \sin t, t \rangle$  (D)  $\vec{r}(t) = \langle t, t^2, 4 \rangle$  (E)  $\vec{r}(t) = \langle 2 \cos t, 2 \sin t, 2 \rangle$

11. Consider the system of equations:

$$\begin{cases} xu^2v + y \cos(u) = 1 \\ \ln(uv) + x^2y - u + v = 0 \end{cases}$$

Assume that this system defines  $u$  and  $v$  implicitly as functions of  $x$  and  $y$  near the point  $P(x, y, u, v) = (1, 0, 1, 1)$ . Show that the system can be solved for  $u$  and  $v$  in terms of  $x$  and  $y$  near  $P$  and find the value of  $\partial u / \partial y$  at  $P$ .

12. a. Consider the function

$$f(x, y) = \begin{cases} \frac{x^2 + \sin(x) \cos(y)}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}.$$

Find  $f_x(0, 0)$  using the formal definition of the partial derivative.

- b. Evaluate the following limit or show that it does not exist.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y^2}{x^6 + y^4}$$

13. Find the absolute maximum and minimum values of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y$$

on the closed triangular region  $D$  with vertices  $(0, 0)$ ,  $(4, 0)$ , and  $(0, -4)$ . Show all your work, including the analysis of the interior critical points and the boundary of the region.

1. Given cylindrical coordinates  $(r, \theta, z) = (\sqrt{3}, -\frac{\pi}{4}, -1)$ . Using conversion formulas:

$$\begin{aligned}\rho &= \sqrt{r^2 + z^2} = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3 + 1} = 2 \\ \theta &= -\frac{\pi}{4} \\ \phi &= \arccos\left(\frac{z}{\rho}\right) = \arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}\end{aligned}$$

Thus  $(\rho, \theta, \phi) = (2, -\frac{\pi}{4}, \frac{2\pi}{3})$ .

Answer: (A)

2. We can solve this question by trial and error. Inspect  $r = \cos^2 \theta \sin \theta$ .

- When  $\theta = 0, \pi$ ,  $r = 0$ .
- For  $\theta \in (0, \pi)$ ,  $\sin \theta > 0$  and  $\cos^2 \theta \geq 0$ , creating a loop along the positive  $y$ -axis.
- The factor  $\cos^2 \theta$  pinches the loop at  $\theta = \frac{\pi}{2}$ , matching the given shape.

Answer: (A)

3. Given point  $P(1, 2, 3)$  and plane  $x + y + z = 0$  with normal vector  $\vec{n} = \langle 1, 1, 1 \rangle$ .

The orthogonal line through  $P$  is  $\vec{r}(t) = \langle 1 + t, 2 + t, 3 + t \rangle$ . Substitute into the plane equation.

$$(1 + t) + (2 + t) + (3 + t) = 0 \implies 3t + 6 = 0 \implies t = -2$$

Substitute  $t = -2$  back yields the projection point.

$$P' = (1 - 2, 2 - 2, 3 - 2) = (-1, 0, 1)$$

Answer: (A)

4. Find intersection points for  $r_1 = 3 \cos \theta$  and  $r_2 = 1 + \cos \theta$ .

$$3 \cos \theta = 1 + \cos \theta \implies 2 \cos \theta = 1 \implies \theta = \pm \frac{\pi}{3}$$

Calculate area.

$$\begin{aligned}A &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} (r_1^2 - r_2^2) d\theta = \int_0^{\pi/3} (8 \cos^2 \theta - 2 \cos \theta - 1) d\theta \\ &= \int_0^{\pi/3} (3 + 4 \cos 2\theta - 2 \cos \theta) d\theta \\ &= [3\theta + 2 \sin 2\theta - 2 \sin \theta]_0^{\pi/3} = \pi\end{aligned}$$

Answer: (A)

5. Rearrange  $2z^2 - \frac{x^2}{2} - y^2 - 5 = 0$ .

$$\frac{z^2}{5/2} - \frac{x^2}{10} - \frac{y^2}{5} = 1$$

The standard form  $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  has two negative minus signs and one positive sign, representing a hyperboloid of two sheets.

Answer: (A)

6. The direction of maximum increase of  $f(x, y, z) = x^2 + 2y^2 - 3z$  is given by gradient  $\nabla f$ :

$$\nabla f = \langle 2x, 4y, -3 \rangle$$

Evaluate at  $P(1, 1, 2)$ .

$$\nabla f(1, 1, 2) = \langle 2, 4, -3 \rangle$$

Answer: (A)

7. Given  $w = \ln(x^2 + y^2)$ ,  $x = s \cos t$ , and  $y = s + \sin t$ . At  $(s, t) = (1, \frac{\pi}{6})$ :

$$x = \frac{\sqrt{3}}{2}, \quad y = \frac{3}{2}, \quad x^2 + y^2 = 3$$

Applying the Chain Rule  $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t}$ :

$$\begin{aligned} \frac{\partial w}{\partial x} &= \frac{2x}{x^2 + y^2} = \frac{\sqrt{3}}{3}, & \frac{\partial x}{\partial t} &= -s \sin t = -\frac{1}{2} \\ \frac{\partial w}{\partial y} &= \frac{2y}{x^2 + y^2} = 1, & \frac{\partial y}{\partial t} &= \cos t = \frac{\sqrt{3}}{2} \end{aligned}$$

$$\frac{\partial w}{\partial t} = \left( \frac{\sqrt{3}}{3} \right) \left( -\frac{1}{2} \right) + (1) \left( \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{3}$$

Answer: (E)

8. For  $y = \frac{x^2}{2}$  and  $z = \frac{x^3}{6}$ , derivatives are  $\frac{dy}{dx} = x$  and  $\frac{dz}{dx} = \frac{x^2}{2}$ .

$$\begin{aligned} L &= \int_0^2 \sqrt{1 + \left( \frac{dy}{dx} \right)^2 + \left( \frac{dz}{dx} \right)^2} dx = \int_0^2 \sqrt{1 + x^2 + \frac{x^4}{4}} dx \\ &= \int_0^2 \left( 1 + \frac{x^2}{2} \right) dx = \left[ x + \frac{x^3}{6} \right]_0^2 = \frac{10}{3} \end{aligned}$$

Answer: (A)

9. For  $F(x, y, z) = x^2 + 2y^2 - z = 0$  and  $G(x, y, z) = x^2 + (y - 1)^2 - 1 = 0$ , gradients at  $P(1, 1, 3)$  are:

$$\nabla F(1, 1, 3) = \langle 2, 4, -1 \rangle, \quad \nabla G(1, 1, 3) = \langle 2, 0, 0 \rangle$$

Cross product gives the tangent vector.

$$\vec{v} = \nabla F \times \nabla G = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & -1 \\ 2 & 0 & 0 \end{vmatrix} = \langle 0, -2, -8 \rangle$$

Negating yields the parallel vector  $\langle 0, 2, 8 \rangle$ .

Answer: (A)

10. Intersecting  $z = x^2 + y^2$  and  $z = 4$  gives  $x^2 + y^2 = 4$  in the plane  $z = 4$ .

Standard parametrization is

$$\vec{r}(t) = \langle 2 \cos t, 2 \sin t, 4 \rangle$$

Answer: (A)

11. Jacobian matrix with respect to  $(u, v)$  at  $P(1, 0, 1, 1)$ :

$$J = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}_{\text{at } P} = \begin{bmatrix} 2xuv - y \sin u & xu^2 \\ \frac{1}{u} - 1 & \frac{1}{v} + 1 \end{bmatrix}_{\text{at } P} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Since  $\det(J) = 2 \cdot 2 - 1 \cdot 0 = 4 \neq 0$ , the system defines  $u, v$  implicitly near  $P$ . Differentiate with respect to  $y$ .

$$\begin{aligned} 2u \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} + \cos(1) &= 0 \implies 2 \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = -\cos(1) \\ 2 \frac{\partial v}{\partial y} + 1 &= 0 \implies \frac{\partial v}{\partial y} = -\frac{1}{2} \end{aligned}$$

Substitute  $\frac{\partial v}{\partial y}$ .

$$\frac{\partial u}{\partial y} = \frac{1 - 2 \cos(1)}{4}$$

12. a. Use the formal definition of  $f_x(0, 0)$ .

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^2 + \sin(h)}{h} - 1}{h} = \lim_{h \rightarrow 0} \frac{h^2 + \sin(h) - h}{h^2}$$

Use L'Hôpital's rule twice.

$$f_x(0,0) \stackrel{\text{L'H.}}{=} \lim_{h \rightarrow 0} \frac{2h + \cos h - 1}{2h} \stackrel{\text{L'H.}}{=} \lim_{h \rightarrow 0} \frac{2 - \sin h}{2} = \frac{2 - 0}{2} = \boxed{1}$$

b. Approach  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y^2}{x^6 + y^4}$  along paths  $y = kx^{3/2}$ .

$$\lim_{x \rightarrow 0^+} \frac{x^3(k^2 x^3)}{x^6 + k^4 x^6} = \frac{k^2}{1 + k^4}$$

Since the result depends on  $k$ , the limit does not exist.

13. The extreme value theorem states that a continuous function on a specific region will always have its absolute maxima and minima.  $f$  is continuous everywhere on its domain. Check the interior points.

$$\nabla f = \langle 2x - 2, 2y + 4 \rangle = \langle 0, 0 \rangle \implies (1, -2)$$

Check if the point is inside the region.

$$y > x - 4 \longrightarrow 1 - (-2) = 3 < 4$$

$(1, -2)$  is inside. We have  $f(1, -2) = -5$ .

Check the boundary.

- $x = 0$ :  $g(y) = y^2 + 4y \implies y = -2 \implies f(0, -2) = -4$
- $y = 0$ :  $h(x) = x^2 - 2x \implies x = 1 \implies f(1, 0) = -1$
- $y = x - 4$ :  $k(x) = 2x^2 - 6x \implies x = \frac{3}{2}, y = -\frac{5}{2} \implies f\left(\frac{3}{2}, -\frac{5}{2}\right) = -4.5$

Check the endpoints. At endpoints:  $f(0, 0) = 0$ ,  $f(0, -4) = 0$ , and  $f(4, 0) = 8$ .

Compare all the values obtained.

Absolute minimum value is  $-5$  at  $(1, -2)$ . Absolute maximum value is  $8$  at  $(4, 0)$ .